ANWIN JOHN

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SUMMARY

Ph.D. candidate in Chemical Engineering specializing in computational catalysis and electrocatalysis, with expertise in density functional theory (DFT), molecular dynamics, and machine learning for materials discovery. Experienced in developing automated simulation workflows and ML models to study catalytic mechanisms and accelerate materials design.

EDUCATION

University of Wisconsin–Madison (GPA: **3.94**/4.0) — Madison, WI
Doctor of Philosophy in Chemical Engineering — May 2027

Indian Institute of Science (GPA: **9.3**/10.0) — Bengaluru, India
Master of Technology in Chemical Engineering — May 2022

Government Engineering College Thrissur (GPA: **8.86**/10.0) — Thrissur, India
Bachelor of Technology in Chemical Engineering — May 2019

RESEARCH EXPERIENCE

University of Wisconsin-Madison — Madison, WI
PAL-DIRAC: Automated DFT- MLIP Active Learning Framework — January 2026 – present

- Developed PAL-DIRAC, an automated active-learning framework integrating DFT, machine-learned interatomic potentials, ensemble uncertainty, structure sampling/curation, with automated HPC workflows. (https://github.com/anwinjohn98/al-dirac)

University of Wisconsin-Madison — Madison, WI
Predicting Surface Work Functions with Graph Neural Networks — December 2025

- Designed and implemented a custom graph neural network architecture to predict surface work functions from atomistic structures of metal catalysts with adsorbate-electrolyte interfaces.

Influence of solvation on faceted surfaces — August 2025 – present
Advisor: Prof. Jeffrey Greeley

- Develop and apply MLIPs with molecular dynamics (ML-MD) to investigate long-timescale solvation effects on adsorbates on faceted surfaces under constant-voltage conditions.

Purdue University — West Lafayette, IN
Identifying Linear Trends in Electrocatalysis — January 2023 – present
Advisor: Prof. Jeffrey Greeley

- **Automated Catalyst Design & Simulation**: Developed VASP workflows using Python, Bash, ASE, and pymatgen to build transition-metal–water bilayer catalyst models, calculate key thermochemical and electrochemical descriptors, and orchestrate DFT simulations - reducing setup time.
- **Mechanistic Modeling:** Integrated DFT calculations and microkinetic modeling to quantify mechanistic differences between thermal and electro-catalysis-informing the rational design of electro-catalysts.
- **MLIP Integration:** Leveraged foundational MLIP model integrated with VASP to accelerate molecular simulations while preserving the DFT-level accuracy.

Role of Water as Proton Source for CO_2RR January 2024 – present
Advisor: Prof. Jeffrey Greeley

- Performed Ab-initio molecular dynamics (AIMD) simulations paired with potential of mean force (PMF) method to quantify barriers of CO_2 reduction reaction (CO_2RR) pathways, focusing on the role of water as proton source.

Engineering Platinum Surface Facets for Enhanced O_2RR Activity November 2024 – present
Advisor: Prof. Zhenhua Zheng

- Systematically investigated the impact of surface strain and facet orientation on oxygen reduction kinetics, to create useful guidelines for the design of metal nanoparticles.

Indian Institute of Science Bengaluru, India
Discovering Single-Atom Catalyst for CO_2RR August 2021 – July 2022
Advisor: Prof. Ananth Govind Rajan

- Collaborated with ISRO to perform DFT simulations on CO_2 reduction over Cu-based single-atom alloys, identifying Al–Cu as the most active catalyst. The European Society Journal for Catalysis: ChemCatChem

Kinetic Monte Carlo(KMC) for Adsorption-Desorption December 2021

- Developed a KMC code to model the non-ideal adsorption-desorption of two distinct mono-dentate adsorbates, incorporating for adsorbate-adsorbate interaction, and adsorbate-site interaction treated differently for each adsorbate.

MENTORSHIP AND MANAGMENT EXPERIENCE

Purdue University West Lafayette, IN
Teaching Assistant - Ph.D. Level Chemical Engineering Courses
Applied Mathematics for Chemical Engineers, Advanced Chemical Engineering Thermodynamics, Advanced Modeling for Catalysis Studies

- Assisted in course preparation, delivered lectures, and facilitated problem-solving sessions for doctoral students.
- Assisted with installation and configuration of computational tools, guiding students in preparing simulation input files and developing basic automation workflows.

HPC Software and Workflow Management January 2024 – Present
Manage installation, and configuration of molecular simulation software and Python packages on HPC clusters; develop alias scripts to streamline module access, enabling efficient, and seamless use of computational tool across the research group.

PUBLICATIONS

- **John, A**., Verma, A. M., Shaneeth, M., & Govind Rajan, A. (2023). Discovering a Single-Atom Catalyst for CO2 Electroreduction to C1 Hydrocarbons: Thermodynamics and Kinetics on Aluminum-Doped Copper. ChemCatChem, 15(14). https://doi.org/10.1002/cctc.202300188
- **John, A**., Greeley, J. Understanding the difference between thermal and electro-catalysis for CHx species on transition metals. (in-preparation).
- **John, A**., Greeley, J. Structurally induced deviations from Brønsted-Evans-Polanyi relationships in electrocatalytic proton-coupled electron transfer. (in-preparation).
- Morankar, A. R., **John, A.,** Deshpande, S., & Greeley, J. Investigating water as a proton source for HER vs. CO2R under alkaline pH conditions (in-preparation).
- **John, A.,** Greeley, J., Zeng, Z. Elucidating bunching and its impact on oxygen reduction reaction activity on Pt facets (in-preparation).

PRESENTATIONS

Oral talk: 2026 AIChE Annual Meeting, 29th North American Catalysis Society Meeting 2025, 248th ECS Meeting 2025, 2025 AIChE Annual Meeting
Posters: CCC Symposium 2026, SUNCAT Summer Institute 2023

SKILLS

Programming: Python (ASE, Pymatgen, networkx), MATLAB, Bash
Computational tools: VASP, VASPsol, vasp-interactive, Gaussian, LAMMPS, MACE, fairchem
Scientific areas: Catalyst design, electrochemistry, surface science, statistical mechanics, material discovery, applied mathematics